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Paper Code : CS-605A

OPERATIONS RESEARCH

Time Allotted: 3 Hours | *Full Marks*: 70

The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable.

Group – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for *any ten* of the following: 1×10=10

(i) Given a system of m simultaneous equations in n unknowns ($m < n$), the number of basic variables will be

(a) m (b) n

(c) $m-n$ (d) None of these

(ii) If some basic variables are zero, then the solution to the corresponding LPP is called

(a) Non-degenerate (b) Optimal

(c) Non-optimal (d) Degenerate

(iii) If any of the constraints in the primal problem be a perfact equality, then the corresponding dual variable is

(a) always positive (b) always negative

(c) equal to zero (d) unrestricted in sign

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(iv) Which of the following is not a method to obtain the initial basic feasible solution in transportation problem?

(a) VAM
(b) Least cost method
(c) North-West corner method
(d) MODI method

(v) When maximin and minimax values of the game are same, then

(a) these is a saddle point
(b) solution does not exist
(c) strategies are mixed
(d) None of these

(vi) An LPP having an artificial variable at positive level in the basis, when all $z_j - c_j \geq 0$ has

(a) unbounded solution
(b) basic feasible solution
(c) infeasible solution
(d) alternative optimal solution

(vii) Traffic intensity of a simple queue is given by

(a) $\rho = \frac{\mu}{\lambda}$
(b) $\rho = \frac{\mu t}{\lambda}$
(c) $\rho = \frac{\mu}{\lambda t}$
(d) $\rho = \frac{\lambda}{\mu}$

(viii) If the dual has an unbounded solution, then primal has

(a) an unbounded solution
(b) an infeasible solution
(c) a feasible solution
(d) None of these

(ix) The shortest route between two given nodes in a Network is determined by the following:

(a) Dijkastra's Algorithm
(b) Floyd's Algorithm
(c) Critical Path Method
(d) None of these

(x) What is the method used to solve an LPP involving artificial variables?

(a) Simplex method
(b) Charne's Big-M method
(c) VAM
(d) None of these

(xi) In PERT, the expected variance (σ^2) of each activity is given by

(a) $\left(\frac{t_e - t_o}{6}\right)^2$
(b) $\left(\frac{t_m - t_o}{6}\right)^2$
(c) $\left(\frac{t_m - t_p}{6}\right)^2$
(d) $\left(\frac{t_o + 4t_m + t_p}{6}\right)$

where, t_o = Optimistic time
t_p = Pessimistic time
t_m = Most likely time

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(xii) If there were 'n' workers and 'n' jobs, there would be

(a) n solutions (b) $n!$ solutions

(c) $(n-1)!$ solutions (d) n^n solutions

Group – B

(Short Answer Type Questions)

Answer *any three* of the following: 5×3=15

2. Using graphical method solve the following LPP:

$$\text{Maximize} \quad Z = 20x_1 + 10x_2$$

$$\text{Subject to} \quad x_1 + 2x_2 \leq 40$$

$$3x_1 + x_2 \geq 30$$

$$4x_1 + 3x_2 \geq 60;$$

$$x_1, x_2 \geq 0.$$

3. Define Convex set. Show that the set given by $x = \{(x_1, x_2), 4x_1^2 + 9x_2^2 \leq 36\}$ is a convex set.

4. Find the expected number of customers waiting in the queuing system L_q, when the queue is in the steady state corresponding to the model (M/M/1) : (∞/FCFS).

5. For what value of λ, the game with the following pay-off matrix is strictly determinable?

		Player B		
		B_1	B_2	B_3
Player A	A_1	λ	5	2
	A_2	-1	λ	-8
	A_3	-2	3	λ

6. Write the dual of the following primal problem:

$$\text{Maximize} \quad z = 3x_1 + 4x_2 - x_3$$

$$\text{Subject to} \quad x_1 + 2x_2 + x_3 \leq 5$$

$$-8x_1 + 6x_2 + x_3 = 19$$

$$3x_1 + x_2 - x_3 \geq 7$$

$$x_1, x_2 \geq 0$$

and x_3 is unrestricted in sign.

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Group – C

(Long Answer Type Questions)

Answer *any three* of the following. 15×3=45

7. (a) Using simplex algorithm solve the following LPP:

Minimize $Z = 2x_1 + 3x_2$

Subject to $3x_1 + 5x_2 \geq 30$

$5x_1 + 3x_2 \geq 60$

$x_1, x_2 \geq 0$.

(b) Solve the following LPP by Charne's Big-M method:

Maximize $Z = 2x_1 + x_2 - 3x_3$

Subject to $x_1 + x_2 + 2x_3 \leq 5$

$2x_1 + 3x_2 + 4x_3 = 12$,

$x_1, x_2, x_3 \geq 0$. 6+9=15

8. (a) Solve the transportation problem and checking the optimality, find the optimal solution:

	D_1	D_2	D_3	Supply
O_1	4	3	2	10
O_2	1	5	0	13
O_3	3	8	5	12
Demand	8	5	4	

(b) Find the optimal assignment and minimum cost for the assignment with the following cost matrix:

	I	II	III	IV	V
A	6	5	8	11	16
B	1	13	16	1	10
C	16	11	8	8	8
D	9	14	12	10	16
E	10	13	11	8	16

7+8=15

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9. (a) Use dominance rule to reduce the pay-off matrix and hence solve the game with the following pay-off matrix:

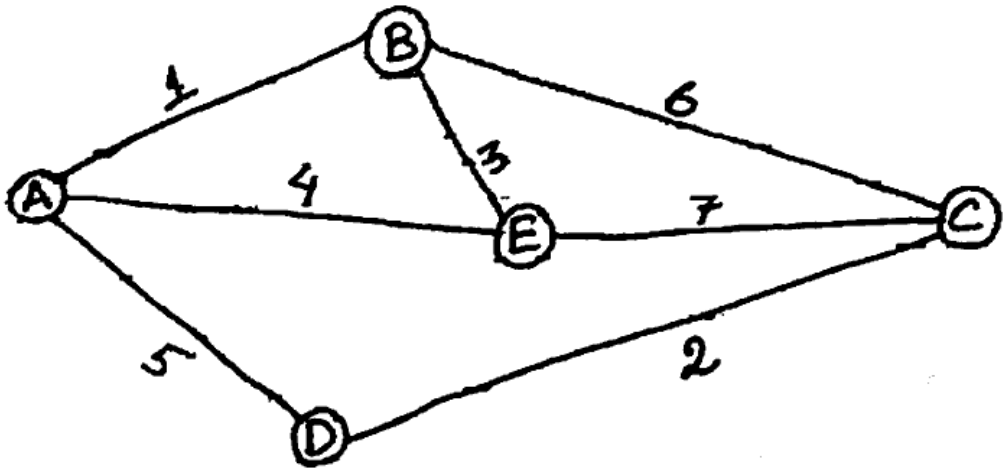
	Player B			
Player A	5	−10	9	0
	6	7	8	1
	8	7	15	1
	3	4	−1	4

(b) Solve the following game by graphical method:

	Player B			
Player A	1	0	4	1
	−1	1	−2	5

8+7=15

10. (a) Use Floyd's algorithm to determine a shortest path and shortest distance from A to C for the following network:



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(b) A project has the following time schedule:

Activity	Time in weeks
1–2	4
1–3	1
2–4	1
3–4	1
3–5	6
4–9	5
5–6	4
5–7	8
6–8	1
7–8	2
8–9	1
8–10	8
9–10	7

Construct network and find critical path and its duration. 7+(3+3+2)=15

11. (a) A small project is composed of seven activities whose time estimates are given in the following table :

Activity		Time required (days)		
Event	Name	t_o	t_m	t_p
1–2	A	6	6	24
1–3	B	6	12	18
1–4	C	12	12	30
2–5	D	6	6	6
3–5	E	12	30	48
4–6	F	12	30	42
5–6	G	18	30	54

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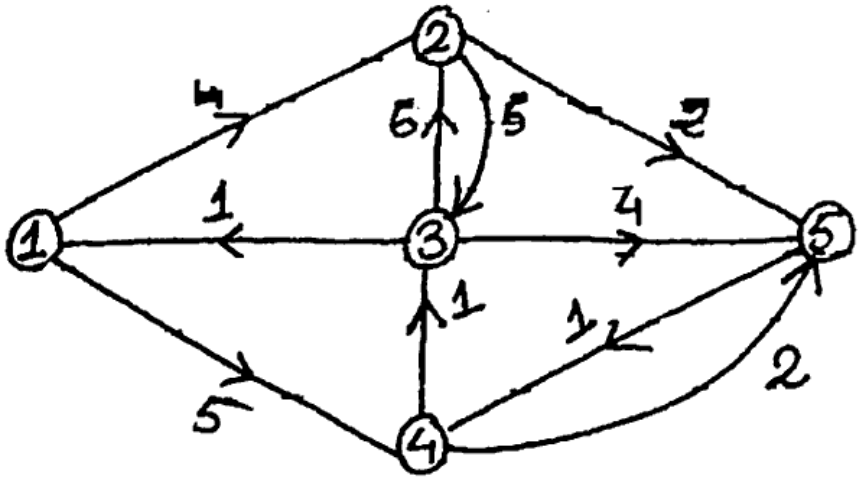
(i) Find the expected duration and variance for each activity.

(ii) What is the expected project length?

(iii) Calculate the variance and standard derivation of the project length. 4+3+2

(b) Find the value of maximum flow for the following network by Ford-Fulkerson algorithm.

4+3+2+6=15



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